

Simarpreet Singh

Student Developer

Portfolio:
<https://www.simarpreetsingh.org>
GitHub:
<https://github.com/Sim20004>
Email: me@simarpreetsingh.org

Education Queen Mary’s Grammar School, Walsall Year 9 Student	Personal Profile Student developer with experience in Python, Rust, web development, Linux administration, and self-hosting. Creator of several software projects including a privacy-focused chat platform, a programming language, and an AI-assisted music learning system. Passionate about open source software and learning new technologies.
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Technical Skills	
Programming	Technologies
• Python • Rust • HTML • CSS • Java	• Linux • Git • GitHub • Docker • Nginx • Cloudflare

In practice: Rust builds the Teapot compiler; Python and Linux run the self-hosted infrastructure behind Vlyra AI; HTML/CSS shape the portfolio site; Git, GitHub, Docker, Nginx, and Cloudflare handle version control and deployment across all of the projects below.

Qualifications & Achievements		
Computer Science	Music	Mandarin
OCR Entry Level Certificate in Computer Science (ELCS) 30/07/2026	ABRSM Grades 1-4 Piano (Practical) with Distinction, Merit, Distinction, and Merit respectively 2023-2026 Trinity College Grade 4 Clarinet (Performance) with Merit 12/02/2026	National Achievement Hanyu Qiao Zhongwen Shuo Bisai (China Bridge National Mandarin Speaking Competition) organised by Hanyu Qiao Organisation (China Bridge) First place in Primary Category nationwide. 26/10/2025

Projects - repositories are linked in portfolio above.

FOSSil Chat *(Ongoing)*

- Privacy-focused chat platform

TeapotOS *(Ongoing)*

- Wrote an operating system from scratch in Assembly, aiming to support Windows, Linux, and MacOS binaries

Teapot Programming Language *(Ongoing)*

- Designed and implemented a custom programming language with lexical analyser, parser, and abstract syntax tree

Personal Portfolio Site *(Ongoing)*

- Designed and developed personal website
- Implemented responsive design and SEO optimisation

Technical Experience

- Self-hosted Linux servers
- Docker container deployment and management
- Nginx web server configuration
- Cloudflare DNS and domain management
- Git and GitHub workflows

Leadership and Public Speaking

- Invited to deliver a speech to officials and representatives involved in UK-China relations
- Presented speech in both Mandarin and English

Languages

Punjabi

Native speaker

English

Native speaker

Mandarin

Intermediate proficiency
Participant in the
Mandarin Excellence
Programme (MEP)

Interests

- Software Development
- Open Source Software
- Linux & Self-Hosting
- Programming Language/Compiler Design
- Piano, Clarinet, and Saxophone
- Cybersecurity

References

Available upon request